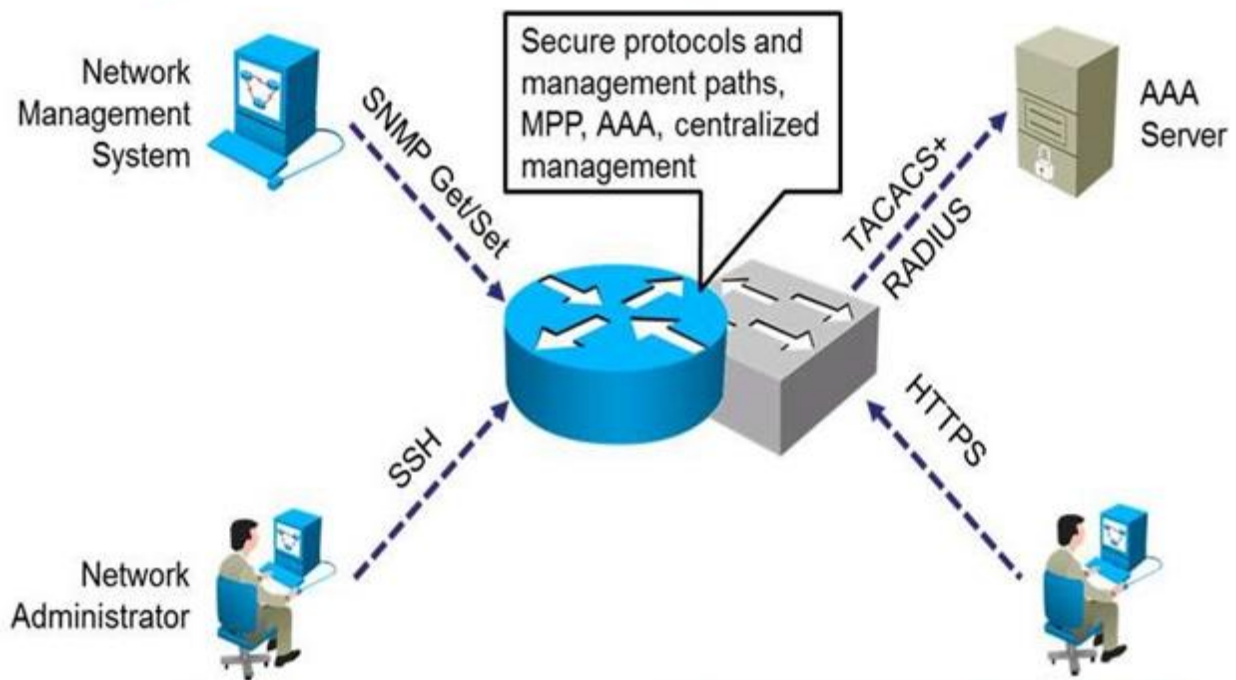


Management Plane.

- o Management Plane traffic is the traffic from the **user to the device**.
- o Protocols & traffic that an admin uses between PC & router or switch itself.
- o For example, using SSH to monitor or configure the Cisco router or Cisco switch.
- o Management plane is used to manage device through its connection to the network.
- o Traffic that flows from administrator's PC when a device is being remotely managed.
- o For security, use AAA, Authenticated NTP, SSH, syslog, SNMPv3 and Parser views.
- o Management Plane protocols are FTP, HTTP, HTTPS, SSH, SNMP, Telnet, TFTP etc.
- o These management protocols are used for monitoring and for CLI access of devices.
- o Management Plane Protection (MPP) is a security feature for Cisco IOS routers.
- o Management Plane Protection (MPP) aka Securing your Management Protocols.
- o MPP feature in Cisco IOS software provides the capability to restrict the interfaces.
- o Greater access control for managing device than allowing management protocols.
- o Restrict interfaces where router permits packets from network management protocols.

Management Plane Security Controls



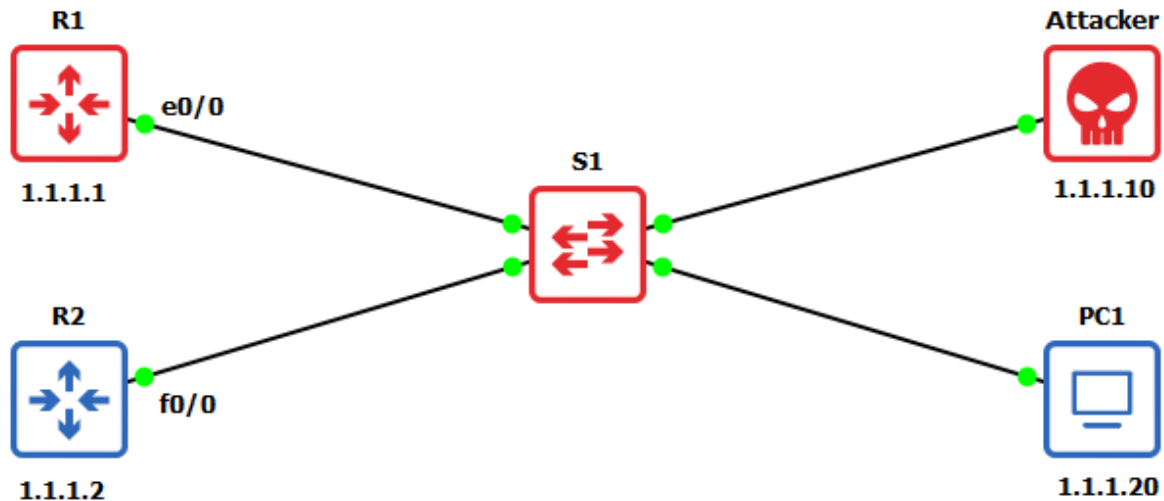
Best Practices for Securing the Management Plane.

- o Enforce password policy on every Network devices such as Router & Switch.
- o Apply the maximum number of login attempts, minimum password length.
- o Implement Role-Based Access Control (RBAC) on Cisco Routers and devices.
- o Implemented Role-Based Access Control through ISE or by CLI parser views.
- o Custom privilege level assignments are an option to restrict privilege levels.
- o Use AAA services to authenticate, authorize & audit any commands of device.
- o Keep accurate time across all network devices using Network Time Protocol.
- o Use encrypted and authenticated versions of SNMP such as SNMP Version 3.
- o Use an OOB method to communicate management traffic between devices.
- o The OOB is a separate VLAN or Interface just for the management traffic only.
- o Enforce the Login Password Retry Lockout policies on every Network devices.
- o Use encrypted management protocols such as SSH, HTTPs and SNMPv3 only.
- o Enable Logging and monitoring on all network devices Routers and Switches.
- o Disable any unnecessary services on network devices Routers and Switches.

Console Port:

- o Every Cisco Router, Cisco Firewall or a Cisco Switch has a console port.
- o Console port also known as the management port on its backside of Device.
- o Console port is used to connect a computer directly to a Cisco router or switch.
- o It manage the router or switch since there is no display device for a router or switch.
- o The console port must be used to initially to install a Cisco Routers or any device.
- o There is no network connection initially to connect using SSH, HTTP or HTTPS etc.
- o Normally Cisco Router, Cisco Switch or Cisco Firewall console port is a RJ45 port.
- o Console port is OOB the management port, which is used by administrators.
- o Console port can be used to log into a router directly without network connection.
- o Console require a terminal emulator application like PuTTY to connect to Router.
- o Console port connect to router when a router cannot be accessed over the network.





Configure Console Authentication R1

Password Only

```

R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
  
```

Username & Password

```

R1(config)# username admin password cisco
R1(config)# line console 0
R1(config-line)# login local
  
```

AAA Local Database

```

R1(config)# aaa new-model
R1(config)# aaa authentication login default local
R1(config)# username admin password 123
R1(config)# line console 0
R1(config-line)# login authentication default
  
```

Enable Mode Passwords

```

R1(config)# enable password cisco
R1(config)# do show running-config | include password
R1(config)# service password-encryption
R1(config)# enable secret cisco
R1(config)# do show running-config | include secret
R1(config)# enable algorithm-type sha256 secret cisco
R1(config)# do show running-config | include secret
R1(config)# enable algorithm-type scrypt secret cisco
R1(config)# do show running-config | include secret
  
```

Brute Force Attack Protection

```
R1(config)# login block-for 120 attempts 5 within 60
```

```
R1(config)# login delay 3
```

```
R1(config)# login on-success log
```

```
R1(config)# login on-failure log
```

```
R1(config)# login quiet-mode access-class Q
```

```
R1(config)# ip access-list standard Q
```

```
R1(config-nsacl)# permit host 1.1.1.20
```

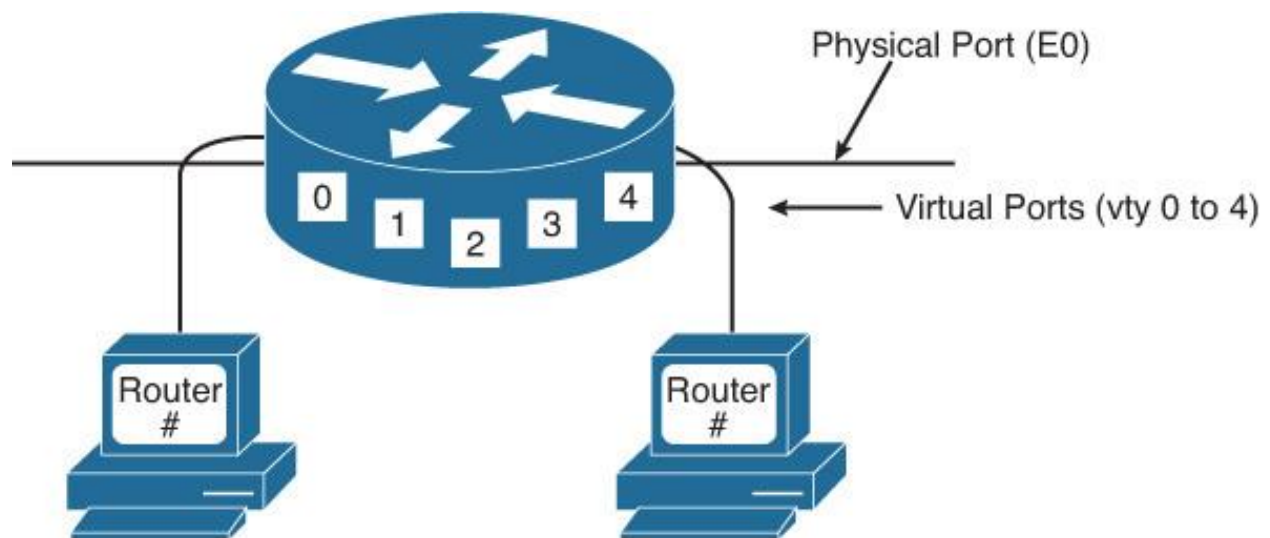
```
R1(config)# security authentication failure rate log
```

```
R1# show login failures
```

```
R1(config)# security passwords min-length 8
```

SSH (Secure Shell):

- o SSH is network term, which stands for **Secure Shell**.
- o SSH provides secure remote access connection to network devices.
- o SSH (Secure Shell) are two versions SSH Version 1 and SSH Version 2.
- o Communication between client & server is encrypted in both SSH version.
- o SSH (Secure Shell), version 2 is more secure than version SSH Version 1.
- o SSH (Secure Shell) commonly uses TCP port 22 to connect to devices.
- o SSH, as the preferred management protocol under the VTY interfaces.
- o SSH provides a secure and reliable mean of connecting to remote devices.
- o SSH (Secure Shell), Version 2 is the more secure and commonly used version.
- o SSH (Secure Shell), require an IOS image that supports crypto features.
- o SSH is a more secure way to configure Routers, Switches or Firewalls.



SSH for Secure Management	
R1(config)# ip domain-name ksa.com	
R1(config)# crypto key generate rsa modulus 1024	
R1(config)# ip ssh timeout 60	
R1(config)# ip ssh authentication-retries 3	
R1(config)#crypto key zeroize rsa	
R1(config)#ip ssh version 2	
R1(config)#username admin privilege 15 secret cisco	
R1(config)#line vty 0 4	
R1(config-line)#login local	
R1(config-line)#transport input ssh	
R1(config-line)# exec-timeout 5 0	
R1#show ssh	R1#show ip ssh
R1#show crypto key mypubkey rsa	R1#debug ip ssh

Allow Specific Hosts for Remote Management	
R1(config)#access-list 1 permit host 1.1.1.20	
R1(config)#line vty 0 4	
R1(config-line)#access-class 1 in	

Disable Auxiliary Port	
R1(config)# line aux 0	R1(config-line)# transport input none
R1(config-line)# no exec	R1(config-line)# exec-timeout 0 0
R1(config-line)# no transport input	R1(config-line)# no transport output

HTTP & HTTPS:

- o HTTP is a web term, which stands for Hyper Text Transfer Protocol.
- o While HTTPS is also web term, stands for Hypertext Transfer Protocol Secure.
- o Cisco IOS has a HTTP server to managed web-based communication with router.
- o HTTP Sends data in clear text over Transmission Control Protocol (TCP) port 80.
- o HTTP can be used for GUI access on Cisco devices such as Routers and Switches.
- o Similar to HTTP, but HTTPS is encrypted over Transmission Control Protocol port 443.
- o Cisco Routers permit a user to connect to a router using Hyper Text Transfer Protocol.
- o Routers permit user to connect to a router using Secure Hyper Text Transfer Protocol.

Web Authentication R1
R1(config)#ip http server
R1(config)#ip http authentication local
R1(config)#ip http max-connections 2
R1(config)#ip http secure-server
R1(config)#ip http authentication aaa
R1(config)#ip http ?
R1(config)# username admin privilege 15 password 123
R1#show ip http client connection
R1# show ip http server connection

Management Plane Protection on R2
R2(config)#control-plane host
R2(config-cp-host)#management-interface fastEthernet 0/0 allow ssh snmp https
R2#show control-plane host features

Auto Secure Routers	
Commands	Descriptions
R1# auto secure	Require User input to secure the router
R1# auto secure no-interact	Without User input secure the router
R1# show auto secure config	View auto secure configuration